Memory Circuits & Systems

Exercise 1
Static Noise Margin of SRAM

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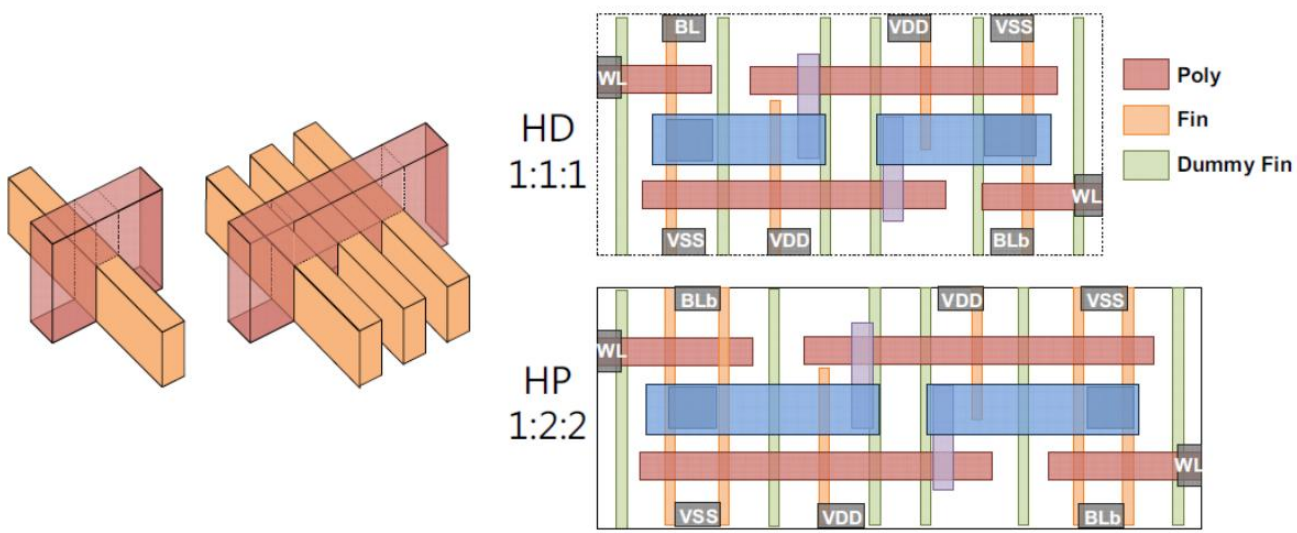
FinFET SRAM

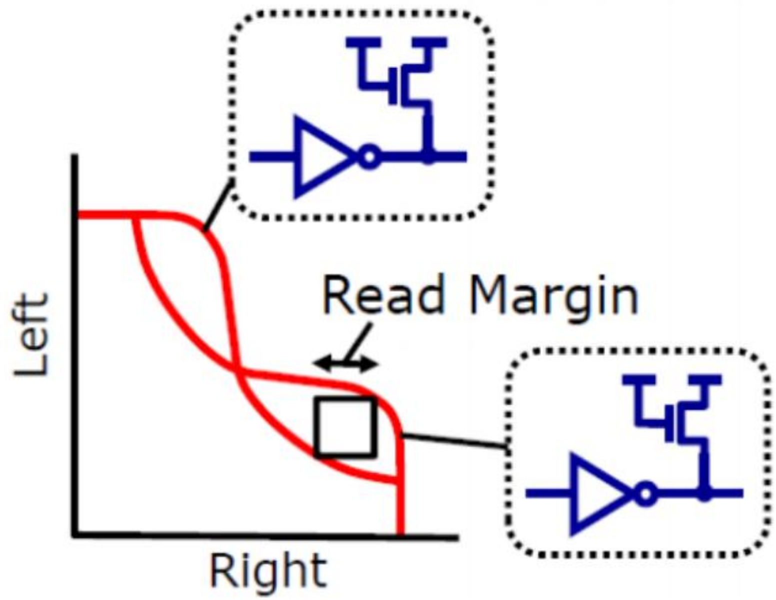
- In FinFET, the width which is given by $W = (T_{FIN} + 2H_{FIN}) \times N_{FIN}$ is quantized.
- For a high-density FinFET 6T-SRAM cell and a high-performance FinFET 6T-SRAM cell, the width ratio of Pull-Up PMOS, pass-gate NMOS and Pull-down NMOS are 1:1:1 and 1:2:2, respectively

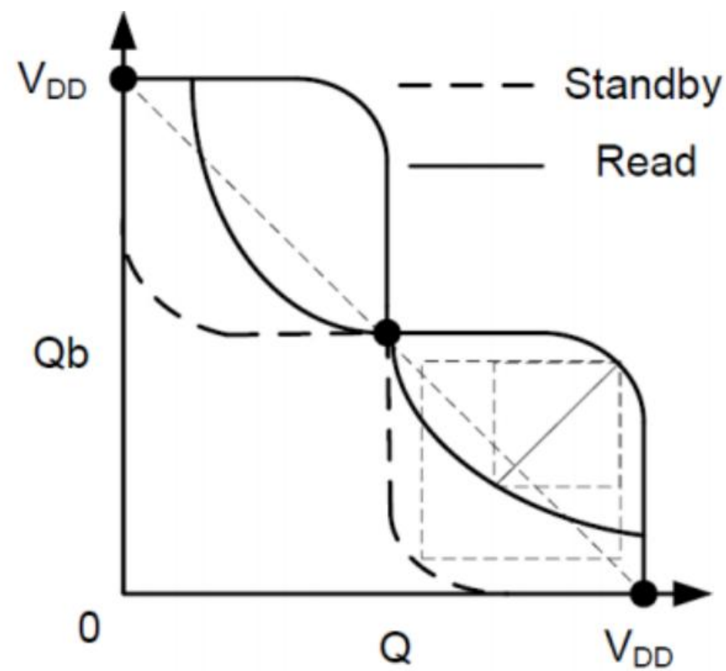


Fin-based SRAM bitcell design

Butterfly Curve of SNM and RSNM

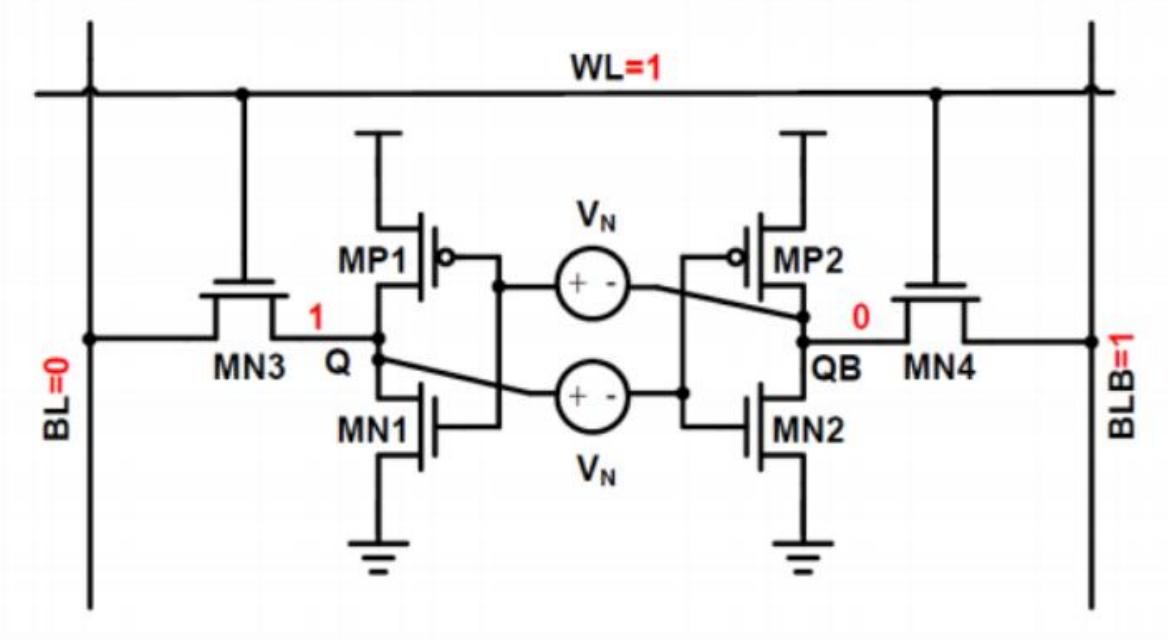
- Using 7nm FinFET model to analyze SNM and RSNM of 1:1:1 and 1:2:2 FinFET SRAM cells
  - Plot butterfly curve of data retention and read modes
    - Hint: DC analysis and get data by .print

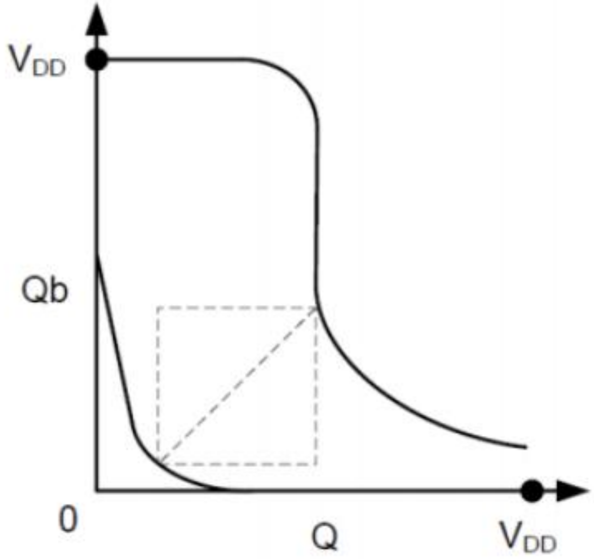




Write Noise Margin (WNM) Curve

- Using 7nm FinFET model to analyze WNM of 1:1:1 and 1:2:2 FinFET SRAM cells
  - Measure the transfer curve of the inverter and plot the curve





Static Noise Margin Analysis

- **Analyze** SNM, RSNM and WNM of 1:1:1 and 1:2:2 FinFET SRAM cells under
  - different supply voltages from 0.7V to 0.4V
    - Bitline/wordline voltages are the same as that of cell voltage
  - different wordline voltages
    - 0.9V to 0.5V when VDD =0.7V

Post-Simulation of SRAM Cell

- **Extract parasitic RC** from the TA-provided SRAM layout (1:1:1). Perform post-simulation using this extracted model to measure the **SNM, RSNM, and WNM**. Compare these results with your pre-simulation data.
  - Measure the SNM, RSNM, and WNM under
    - different supply voltages from 0.7V to 0.4V
      - Bitline/Wordline voltages are the same as that of the cell voltage
    - different Wordline voltages
      - 0.9V to 0.5V when VDD =0.7V
- **Report Requirement:**
  - Explain how you modified the extracted netlist to perform these measurements

Report Requirements

- **Required Waveforms:**
  - Pre-sim (1:1:1, 1:2:2), Post-sim (1:1:1)
    - SNM (fixed, sweep VDD)
    - RSNM (fixed, sweep VDD, WL)
    - WNM (fixed, sweep VDD, WL)
- **Analysis:**
  - Your report must include detailed discussions and observations based on the simulation results
- **Grading Policy:**
  - Pre-sim (1:1:1) - **30 points**
  - Pre-sim (1:2:2) - **30 points**
  - Post-sim (1:1:1) - **40 points**

Submission on e3 platform

- Please compress your **report** & **source codes in a single compressed file (.zip)** and upload this single file on E3 platform
  - The modified extracted netlists should be provided
- Naming rules of the compressed file
  - Upload file: **Ex1_#ID.zip (including report & source code)**
  - **e.g. Ex1_315123123.zip**
- Due date: 4/7 PM 23:55